

# Introduction to Traffic Flow Theory

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The course is called Traffic Flow Theory (TFT).

What is traffic flow theory?

- **Traffic**: **movement** of vehicles, ships, persons, etc., in an area, along a street, through an air lane, over a water route, etc.
- **Flow**: to move along in a **stream**
  - Study on vehicles moving along in a traffic stream
- **Theory**: a body of principles, theorems
  - To produce a **mathematical** theory

- Due to the individual reactions of **human** drivers, vehicles do not interact simply following the laws of mechanics.
- Thus, traffic phenomena are complex and **nonlinear**, depending on the interactions of a large number of vehicles.

- No satisfactory general theory that can be consistently applied to real-world traffic flow
- Current traffic models use a mixture of empirical and theoretical techniques.
- These models are then developed into understanding phenomena and forecasts,

## **The ultimate goal of science is to**

uncover the underlying *laws that govern the universe*, allowing for (1) a deeper comprehension of our surroundings, (2) predictions, (3) technological advancements.

- No satisfactory general theory that can be consistently applied to real-world traffic flow
- Current traffic models use a mixture of empirical and theoretical techniques.
- These models are then developed into understanding phenomena and forecasts,
  - to explain traffic dynamics, mechanism of driving behavior, and human (driver) behavior,
  - to take account of proposed local or major changes, such as increased vehicle use, changes in land use or changes in mode of transport,
  - to identify areas of congestion where the network needs to be adjusted,
  - etc.



## Traffic Flow Theory = Toolbox

It contains a variety of  
important, practical, and rigorous  
analytical tools  
for traffic research and practice

*Essential for your future career in Civil Engineering*

In the discipline of transportation engineering, it is  
the ONLY course whose name includes the word “theory”.

HARDEST??

partially because of mathematics and abstraction, but

INTERESTING & USEFUL!

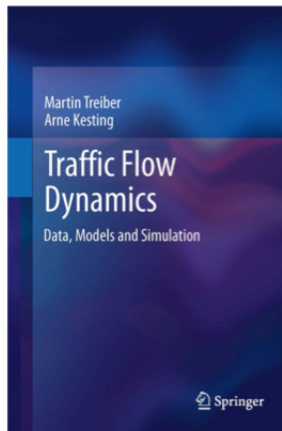
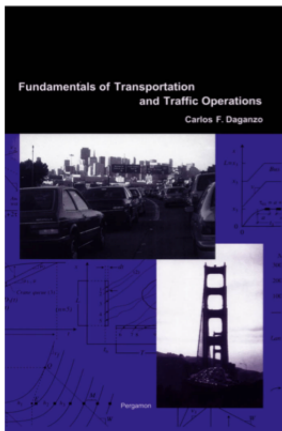
- Trajectories and microscopic traffic variables
- Macroscopic traffic variables
- Important traffic characteristics
- Kinematic wave theory
- Cumulative count curve and queueing analysis
- Cellular automata
- Car-following models, Basic
- Car-following models, Gipps model
- Car-following models, Newell model
- Car-following models, Data-driven model
- Lane-changing models, Basic
- Lane-changing models, Data-driven model
- Intersection control, Basic
- Intersection control, Future
- etc.



# Final Score

- Attendance
- Homework
- Final project

# Recommended Books



Thank you!